

(b) In this example, the bombarding particle is d (deuterium, ${}^2_1\text{H}$), which is an isotope of hydrogen with charge 1 and mass 2. The ejecting particle is alpha (${}^4_2\alpha$) with charge 2 and mass 4. Hence, the balanced equation for this nuclear reaction is



Check: the sum of atomic numbers and mass numbers are the same on both sides of the equation, as they should be.

There is, however, one requirement in this technique. The bombarding particle must possess high kinetic energy (high velocity), otherwise it will not split the target nucleus. The particles are accelerated in a device known as **particle accelerator**. There are two kinds of particle accelerators; (a) cyclotron, and (b) linear accelerator. You may refer to any general chemistry textbook for further information on particle accelerators.

The Nobel Prize in Chemistry 1908

"for his investigations into the disintegration of the elements, and the chemistry of radioactive substances"



Ernest Rutherford

Taken from http://nobelprize.org/nobel_prizes/chemistry/laureates/1908/.